P0B3B-00、 P0DE6-F1、 P0DE6-F2、 P0DE7-F2

Fault diagnosis code
P0B3B-00：Cell voltage sampling failure
P0DE6-F1：Low cell voltage2level failure
P0DE6-F2：Low cell voltage3level failure
P0DE7-F2：High single voltage3level failure
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• BMS fault • BMS balance function fault • Battery pack damage
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P0B3B-00
Single-cell voltage sampling fault (single-cell sampling line falls off)
P0DE6-F1
Unit voltage is lower than the protection threshold: Vmin < 2.5V
P0DE6-F2
Unit voltage is lower than the protection threshold: Vmin < 2.3V
P0DE7-F2
Unit voltage is higher than the safe threshold：Vmax > 3.7V
To set the effect of a fault code condition
Cell function is limited

Troubleshooting steps	
Step 1	General inspection.
	1 Check battery capacity. 2 Check the harness plug of BMS for damage, poor contact, aging and looseness. 3 Check the harness plug of battery pack high voltage for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 2	Check fault codes.
	1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the BMS with diagnostic apparatus. 4 Is there any other fault code except for P0B3B-00、P0DE6-F1、P0DE6-F2、P0DE7-F2?
	YES → Refer to:DTC Summary list(BMS).
NO ↓	
Step 3	Carry out automatic balance operation of battery pack cell.
	1 According to the maintenance process of the manufacturer, the battery pack cells is automatically balanced. 2 Clear the BMS Fault code and re-use the diagnostic instrument to diagnose BMS. 3 Test whether the faulted code is existed.
	YES → Diagnosis end.
NO ↓	
Step 4	Check the connecting cable of BMS.
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Disconnect the harness plug battery pack high voltage. 4 Disconnect the high voltage harness plug of BMS. 5 Check whether the BMS connection cable is rotten, damaged, and water enters. 6 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.

YES ↓		
Step 5	Perform initialization.	
		1 Turn the ignition switch to OFF. 2 Connect the fault diagnosis instrument to the diagnosis interface. 3 Turn the ignition switch to ON. 4 The BMS module was initialized with the diagnostic instrument. 5 Can the BMS be initialized and is the BMS restored?
		NO → Repair or change the BMS module.
YES ↓		
Step 6	Test whether BMS is norm.	
		1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug CH1001 of BMS. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BMS plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
		YES → Replace the BMS.
NO ↓		
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.		
Next Step ↓		
Diagnosis end.		